Rasmus Frigaard Lemvig - Curriculum Vitae

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Education

University of Copenhagen
PhD student. Principal supervisor: Martin Bladt. Co-supervisor: Mogens Steffensen *(May 2025 - present)*
Working thesis title: *Multi-state models: modern methods, machine learning and meditations* $((MM)^3)$

University of Copenhagen
Master of Science in Actuarial Mathematics *(Sep 2023 - Jan 2025)*
Thesis: Machine learning methods for survival and multi-state models
Two-part thesis on reserving under stress scenarios in life insurance and insurance pricing uisng GLMs and machine learning.

University of Copenhagen
Bachelor of Science in Actuarial Mathematics *(Sep 2021 - Jun 2023)*

University of Copenhagen
Bachelor of Science in Mathematics *(Sep 2018 - Jun 2021)*
Thesis: *Cubic and Quartic Reciprocity*

Experience

University of Copenhagen - Department of Mathematical Sciences
PhD student *(May 2025 - present)*. PhD student in the section of Insurance and Economics (IE). Interests include the mathematics and statistics of life insurance, in particular multi-state modelling, and their intersection with machine learning. Other interests include mathematical aspects of risk management and finance.

Scientific assistant *(Feb 2025 - May 2025)*

Teaching assistant *(Sep 2020 - present)* 24 times in total, spanning 20 different courses. Noteworthy courses include:

- Statistics for Insurance
- Quantitative Risk Management
- Topics in Probability
- Basic Life Insurance Mathematics
- Statistical Methods
- Brownian Motion
- Continuous Time Finance
- Advanced Probability Theory 2
- Stochastic Processes in Non-Life Insurance
- Insurance and Law
- Probability Theory 2

- Probability Theory

My complete teaching history can be found here: https://rasmusfl.github.io/work.html.

Manuscripts and projects

1. *Random Jump Forests*
   Martin Bladt and Rasmus Frigaard Lemvig
2. *Local estimation of transition rates of jump processes through discretization*
   Martin Bladt and Rasmus Frigaard Lemvig
3. *Consistency of Honest Decision Trees and Random Forests*
   Martin Bladt and Rasmus Frigaard Lemvig
4. *JumpForests: A random forest R package for jump processes*
   Martin Bladt and Rasmus Frigaard Lemvig
5. Lecture notes for the course *Quantitative Risk Management (QRM)*
   Rasmus Frigaard Lemvig
6. Lecture notes for the course *Stochastic Processes in Non-Life Insurance (SkadeStok) 2023/2024*
   Rasmus Frigaard Lemvig

Programming proficiencies

Experience with programming in **R, Julia, Python/Cython and C++**, including R package development with **RCpp**.

Volunteer work

Danish Youth Association of Science (UNF)
Volunteer *(Jan 2016 - present).* Some highlights include:

- **Cofounder of, coordinator and teacher at** UNF Science Weekend (2019 - 2023), a biannual weekend camp for elementary school students with 85 participants and 30 volunteers.
- **Teacher at** Mathematics Camp from 2021 to 2025 and at Machine Learning Camp 2024.
- **Teacher at** Week in mathematical modelling 2025, a camp with focus on differential equations and their applications. Teacher and coordinator in 2026.
- **Teacher** in several workshops. Topics: Linear algebra, calculus, number theory, graph theory, sums and sequences, differential equations.
- **Coordinator of and teacher in** the project Evenings of Natural Science Education where we taught young Ukrainian refugees.

A detailed description of my work for UNF can be found here: https://rasmusfl.github.io/volunteer.html.

University of Copenhagen - Department of Mathematical Sciences
Student representative in the Board of Education *(Jun 2022 - Jan 2025)*